

**Instructions:** Use proper notation, and show all of your work to receive credit. Round numeric answers to at least 2 decimal places.

1. (4 pts.) Consider a target parameter,  $\theta$ , and corresponding point estimator,  $\hat{\theta}$ .

- (a) (2 pts.) Give the formula for the bias of  $\hat{\theta}$ .

**Solution:** Section 8.2 lecture notes, Definition.

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta$$

- (b) (2 pts.) Give the formula for the error of estimation,  $\epsilon$ .

**Solution:** Section 8.4 lecture notes, Definition.

$$\epsilon = |\hat{\theta} - \theta|$$

2. (4 pts.) Suppose that  $n = 20$  observations are to be taken on  $\ln(\text{LC50})$  measurements and that  $\sigma^2 = 1.4$ . Let  $S^2$  denote the sample variance of the 20 measurements. Find a number  $a$  such that  $P(a \leq S^2) = .975$ .

**Solution:** Homework problem 7.21b.

$$\begin{aligned} P(a \leq S^2) = .975 &\implies P(S^2 \geq a) = .975 \\ &\implies P\left(\frac{(n-1)S^2}{\sigma^2} \geq \frac{19a}{1.4}\right) = .975 \\ &\implies P\left(\chi_{19}^2 \geq \frac{19a}{1.4}\right) = .975 \\ &\implies \frac{19a}{1.4} = \chi_{.975}^2 \text{ (with 19 df)} = 8.90655 \\ &\implies a \approx .6563 \end{aligned}$$

3. (4 pts.) Suppose  $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Gamma}(\alpha, \beta = 3)$ . For the target parameter,  $\alpha$ , calculate  $MSE(\bar{Y})$ .

**Solution:** Section 8.2 lecture notes, Example 1b.

$$V(\bar{Y}) = V\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) = \frac{1}{n^2} \sum_{i=1}^n V(Y_i) = \frac{9\alpha}{n}$$

$$B(\bar{Y}) = E(\bar{Y}) - \alpha = 3\alpha - \alpha = 2\alpha$$

Therefore,

$$MSE(\bar{Y}) = V(\bar{Y}) + [B(\bar{Y})]^2 = \frac{9\alpha}{n} + (2\alpha)^2 = \frac{9\alpha}{n} + 4\alpha^2.$$

4. (3 pts.) A forester studying the effects of fertilization on certain pine forests in the Southeast is interested in estimating the average basal area of pine trees. In studying basal areas of similar trees for many years, he has discovered that these measurements (in square inches) are normally distributed with standard deviation approximately 4 square inches. If the forester samples  $n = 9$  trees, find the probability that the sample mean will be within 2 square inches of the population mean.

**Solution:** Homework problem 7.11.

$$\begin{aligned} P(|\bar{Y} - \mu| \leq 2) &= P\left(\frac{-2}{4/\sqrt{9}} \leq Z \leq \frac{2}{4/\sqrt{9}}\right) \\ &= P(-1.5 \leq Z \leq 1.5) \\ &= \text{normalcdf}(-1.5, 1.5, 0, 1) \\ &\approx .8664 \end{aligned}$$

5. (4 pts.) Let  $Y_1, Y_2, \dots, Y_5$  be a random sample of size 5 from a normal population with mean 0 and variance 1 and let  $\bar{Y} = (1/5) \sum_{i=1}^5 Y_i$ . What is the distribution of  $U = \sum_{i=1}^5 (Y_i - \bar{Y})^2$ ?

**Solution:** Homework problem 7.37b.

$$U = \sum_{i=1}^5 (Y_i - \bar{Y})^2 = 4 \cdot \frac{1}{5-1} \sum_{i=1}^5 (Y_i - \bar{Y})^2 = 4S^2$$

Note that  $n = 5$  and  $\sigma^2 = 1$ , so  $4S^2 = \frac{(n-1)S^2}{\sigma^2}$ . Hence, by Theorem 7.3,  $U \sim \chi_4^2$ .

6. (3 pts.) Let  $Y_1, \dots, Y_n$  be a random sample with  $E(Y_i) = \mu$  and  $V(Y_i) = \sigma^2$ . We showed in class that  $E(S'^2) = \frac{n-1}{n}\sigma^2$ , where  $S'^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$ . Use this fact to show that  $S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$  is an unbiased estimator for  $\sigma^2$ .

**Solution:** Section 8.3 lecture notes, Example 2 (condensed version).

$$E(S^2) = \frac{n}{n-1} E\left[\frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2\right] = \frac{n}{n-1} E(S'^2) = \frac{n}{n-1} \cdot \frac{n-1}{n} \sigma^2 = \sigma^2$$

7. (3 pts.) The reading on a voltage meter connected to a test circuit is uniformly distributed over the interval  $(\theta, \theta + 1)$ , where  $\theta$  is the true but unknown voltage of the circuit. Suppose that  $Y_1, Y_2, \dots, Y_n$  denote a random sample of such readings. Show that  $\bar{Y}$  is a biased estimator of  $\theta$ .

**Solution:** Homework problem 8.12a.

$$E(\bar{Y}) = \frac{1}{n} \sum_{i=1}^n E(Y_i) = \frac{1}{n} \sum_{i=1}^n \frac{\theta + \theta + 1}{2} = \frac{2\theta + 1}{2} = \theta + \frac{1}{2} \neq \theta$$

Hence,  $\bar{Y}$  is biased for  $\theta$ .